

generate high beam currents but at low energies and pulsing rates. Particle accelerators suitable for producing beam weapons need to generate high-intensity and high-energy particle beams.

CPBs are of little use in space. The combined effects of emittance and Coulomb's force of repulsion between like-charged particles broaden the beam. As an example, a 1GeV, 1000-ampere CPB would spread from 1cm to 5m over 1000 km.

Further, the beam is deflected by Earth's magnetic field. By the same study, the 1GeV, 1000-ampere beam would deflect by 1000km over 1000km distance due to Earth's magnetic field. CPBs though can be made to propagate satisfactorily over a few kilometres through an air channel evacuated by heating air in a straight line. Thus, a CPB weapon could be employed for ballistic missile defence. The system could be installed in a few ground based sites in conjunction with either Earth-borne or space-borne radar systems to identify and track incoming ballistic missile warheads.

The CPB weapon could be rapidly pointed at the incoming missile to destroy it. For an interception in air at 10km, an electron beam weapon would typically require 500MeV beam energy and 10,000 amperes of beam current. However, large fixed installations required for CPB weapons, as per current status of technology, may render these vulnerable to sabotage or other forms of attack by an adversary.

**Neutral PBWs.** A NPB weapon consists of neutral atomic particles accelerated to a high kinetic energy level in a particle accelerator. The process of generation of high-energy NPB is as follows:

Hydrogen or deuterium gas is subjected to an enormous electrical charge. The electrical charge produces negatively-charged ions that are accelerated through a long vacuum

tunnel by an electrical potential in the hundreds-of-megavolt range. After the negatively-charged ions have been accelerated, at the end of the tunnel, electrons are stripped from the negative ions, thereby forming the high-speed NPB.

Weapons-class NPB weapons also require energies in hundreds of MeV and beam powers in tens of megawatts. Modern devices have not yet reached this level. Given the state of the art in accelerator technology, achieving the required beam energy and power levels would require hundreds of tons of accelerator hardware and enormous power sources to operate. Due to size, weight, power constraints and inherent complexity, it does not appear feasible that a NPB weapon would see the light of the day before 2025.

NPBs travel in a straight line once these have been accelerated and magnetically pointed just before neutralisation in the accelerator.

Also, an NPB is strongly affected by passage through the atmosphere. It gets attenuated and diffused as it passes through dense gas or suspended aerosols, which makes it far more suitable as compared to a CPB for applications in space against high-flying airborne or space borne targets.

Damage assessment of the target could be possible. When the beam penetrates a target, the target's atomic and subatomic structure produce characteristic emissions that could be used to determine the target's mass or assess the extent of damage to the target.

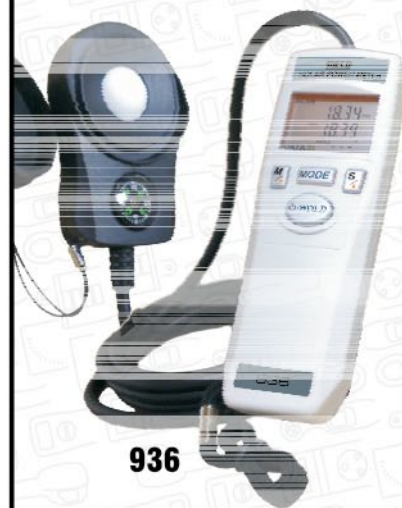
The major disadvantage of a NPB weapon even in space is that it is extremely difficult to sense, which complicates the problem of beam control and direction.

*In the next part we will learn more about PBWs. In subsequent parts we will look at high-power microwaves, less-lethal weapons and high-energy laser weapons.*

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